

Jorge Luis Vargas Lara

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EDUCATION

University of California, Davis

Davis, CA

Master of Science in Water Resources Engineering Sep. 2024 – Dec. 2025

- Bachelor of Science in Civil and Environmental Engineering

Aug. 2019 – June 2024

EXPERIENCE

Intern I

Sept 2025 – Present

Davids Engineering

Davis, CA

- Analyzed regional surface water volumes using Excel pivot tables and Python to identify trends in water usage for SGMA Annual Reports; presented findings via PowerPoint.
- Evaluated contributions of water users, finding irrigation dominates usage while smaller providers have negligible contribution on modeling inputs.
- Investigated discrepancies in IWFWM surface water inputs (2022–2025), identifying transition from EWRIMS to water supplier data as the primary source; authored a professional report recommending the use of supplier data for future reporting.
- Assessed evapotranspiration (ET) inputs for Tehama County using Python and QGIS, determining additional CIMIS stations marginally improve accuracy but do not justify added labor costs. (PowerPoint)
- Conducted fieldwork collecting soil samples to quantify spatial changes in soil moisture and salinity within the Sutter Mutual agricultural region relative to the Sacramento River.

Graduate Researcher

July 2023 – Present

Morales Lab – CEE, University of California-Davis

Davis, CA

- Designed a robust workflow for generating microscale Computational Fluid Dynamics simulations in porous media
- Visualization of preferential flow paths and quantified biofilm-induced hydraulic properties
- Automated simulation processes using Python and Bash, improving reliability and reducing manual labor
- Converted high-resolution experimental biofilm images into 3D models using Python and Blender
- Processed large data sets to be presented in professional quality visuals.
- Presented weekly research updates, demonstrating effective public speaking skills.

Building Supervisor

Sep 2021 – Dec 2023

Memorial Union, University of California-Davis

Davis, CA

- Supervised Information Desk Attendants and served as the primary point of contact for building operations
- Managed table reservations and lost item repository using Excel while delivering high-quality customer service

PROJECTS

Groundwater Model Flow and Transport - Colton, California | *Modflow 6, Flopy (Python), QGIS* May 2025

- Produced comprehensive groundwater model utilize Modflow 6 written in python for reproducibility
- Simulated groundwater flow using public data from monitoring wells and stream gauges to better represent climate
- Processed geospatial data of three plumes to accurately represent contaminated regions using QGIS

Guest Lecturer – Introduction to Turbulence | *Mentoring, Public Speaking, Documentation* December 2025

- Delivered a lecture on numerical fluid simulations, emphasizing model setup, validation, and interpretation.
- Designed simulation template with guidance on installation, scripting, and data analysis.

Diffusion of Oxygen in Multiphase Porous System | *Image Processing, Scripting, Git* January 2026

- Developing numerical simulations building on prior biofilm modeling frameworks to investigate oxygen transport in porous media, emphasizing robust and reproducible methodologies.

TECHNICAL SKILLS

Languages: Python, Matlab, Markdown, Bash, Latex, Spanish

Software Tools: OpenFoam, Modflow 6, Blender, Fusion 360, QGIS, Overleaf, Excel

Developer Tools: Github, Git, Docker, Jupyter Notebook, Visual Studio Code, Spyder, Linux

Libraries: Pandas, NumPy, Matplotlib, Geopandas, Statsmodels, OpenCV, Pyvista

AWARD

Johannes DeVries Graduate Student Award for Outstanding Student in Water Resources Engineering

KEY PUBLICATIONS

- Flow, Mixing, and Reaction Dynamics in Progressively Bioclogged Porous Media - Paper in preparation
- Sensitivity Analysis of Biofilm Permeability in Progressively Bioclogged Porous Media- Paper in preparation

OTHER EXPERIENCES

Roofing Assistant - Seasonal

June 2019 – Present

Valley Roofing

Yucaipa, CA

- Provided customer service by communicating project status and addressing client questions in a professional manner
- Installed shingles and tiles on residential roofing systems using a variety of hand and power tools
- Worked efficiently in physically demanding and extreme weather conditions while maintaining productivity
- Handled and transported large volumes of construction materials and debris, often exceeding hundreds of pounds per project
- Maintained strict adherence to safety protocols while working at heights and in elevated environments

LEADERSHIP

SCIENCE IS ART - CEE ART GALLERY

Founder and organizer of the first annual CEE Art Show, showcasing diverse research within the Civil and Environmental Engineering Department.

RELEVANT COURSEWORK

Turbulence - Water Resources Simulation - Soil Mechanics - Contaminant Fate and Transport in Porous Medium - Groundwater Systems Design - Sediment Transport - Hydrological Time Series Analysis - Deterministic Optimization and Design - Water Resource System Engineering- Jon Herman - Groundwater Modeling- Thomas Harter